

Pre-Registered Confrontation of a $K3 \times T^2$ Dual-Scale Hypothesis with Data: Four Observables, Four Negative Verdicts, and a Common Cause

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Abstract

A research programme built around a “micro/macro dual-scale” reading of string theory on $K3 \times T^2$ identifies the string scale with the geometric mean of the Planck length and the Hubble radius, $\alpha' = \ell_P c/H_0$, and hence a self-dual length $s = \sqrt{\ell_P c/H_0} \approx 47 \mu\text{m}$. We report what happens when every observable this programme can derive — without passing through a compactification that has never been constructed — is written down, frozen by a git tag together with its decision rule *before* the verdict is computed, and then confronted with pinned data. There are exactly four such observables. (P1) s as the radius of one large extra dimension is **excluded** by the Eöt-Wash torsion-balance bound ($< 30 \mu\text{m}$) and fails the neutron-star bound ($< 44 \mu\text{m}$); the programme’s own T-duality fixes the order-one factor that could have rescued it. (C-A) Vacuum energy saturating the Cohen–Kaplan–Nelson bound with infrared length c/H forces a deceleration parameter $q = +1/2$ at every epoch, for every value of its only parameter, and is **excluded** by the observed acceleration. (C-B) Pre-big-bang relic gravitons at the programme’s string scale end at $\approx 6 \times 10^{-5}$ Hz with a peak $\Omega_{\text{GW}} \approx 10^{-65}$, more than fifty orders of magnitude below LISA’s design sensitivity: **not falsifiable in practice**. (C-C) Cosmic F-strings have $G\mu \sim 10^{-62}$: likewise out of reach. The premise shared by C-B and C-C — a meV string scale — was already excluded by LHC resonance searches. The common cause of all four negative verdicts is the identification $\alpha' = \ell_P c/H_0$ itself. Every derivation and every numerical comparison is kernel-checked in Lean 4 (`DualScaleCosmology`, 50 theorems; repository 621 theorems, 0 failing, release `v3.16.1`); the physical identifications are stated as conjectures and the experimental bounds as literature, each pinned to a source file and line range. We record the two freezes made after the relevant bounds were known as retrodictions, and the one made before its comparison document was pinned as a forward prediction. The mathematics of the programme (moonshine, dyons, lattices) is unaffected and is reported separately; the lesson is methodological: pre-registration turned an “order-one tension” into a recorded exclusion and located the single assumption responsible.

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1 Introduction

Papers 7–9 of this series [1–3] built a Lean 4 companion to a “dual-scale” reading of string theory on $K3 \times T^2$ and tested its literal form against the Cohen–Kaplan–Nelson (CKN) bound [5], which it fails by a factor of at least 10^{30} . Paper 9 left one reading open: the self-dual length $s = \sqrt{\ell_P c/H_0} \approx 47 \mu\text{m}$ as the radius of an extra dimension, “disfavoured by an order-one factor, not excluded”. An order-one tension is exactly where a theory can be quietly rescued by a convention chosen after the data are seen. This paper closes that door. It asks two questions, in this order:

1. What in the programme yields a number *with units*, by a derivation that does not pass through an unconstructed compactification (flux quanta, brane wrapping, moduli stabilisation)?
2. For each such number: what does a prediction frozen *before* the verdict — with its decision rule fixed in advance — say, and what do pinned data say about it?

Headline result

Four observables are derivable; all four were frozen and confronted. Two are **excluded** (P1, C-A); two are **out of reach** of any foreseeable experiment (C-B, C-C), and their shared premise is itself excluded. The common cause is the dual-scale identification $\alpha' = \ell_P c/H_0$. The testable forms of the hypothesis do not survive existing data; the programme’s Tier A mathematics does.

Companion material. Lean sources: `DualScaleCosmology/Stream6Verdict.lean`, `Stream7CA.lean`, `Stream7CB.lean`; they build on `SelfDualCutoff.lean`, `DarkEnergyScale.lean`, `CosmicString.lean`, `CKNInstance.lean`. Frozen predictions: `docs/STREAM6_PREDICTION_P1.md`, `docs/STREAM7_PREDICTION_CA.md`, `docs/STREAM7_PREDICTION_CB.md`. Inventory: `docs/STREAM7_HYPOTHESIS_INVENTORY.md`.

2 Epistemic tiers and the pre-registration protocol

Every claim carries one of three labels. **Tier A — Lean kernel**: a proposition checked by the Lean 4 kernel (toolchain v4.33.1, Mathlib), with no `sorry` and with axioms among `propext`, `Classical.choice`, `Quot.sound` (audited per theorem). **Tier L — literature**: a statement taken from the literature, pinned to a hash-registered file in `papers/foundations/` and a line range. **Tier C — conjecture**: a physical identification or hypothesis — never a theorem. A Tier A theorem about numbers with Tier L inputs certifies the arithmetic, not the inputs; a verdict inherits the weakest tier of its premises.

Pre-registration protocol used in this paper

1. **Derive before comparing.** The observable and its numerical value are derived from the programme’s stated assumptions, each labelled measured (M), derived in Lean (D) or assumed (A). A non-empty (A) list makes the prediction Tier C, stated as such.
2. **Freeze by tag.** The statement, the inputs and a *decision rule* with outcome categories are committed and tagged (`stream6-p1-frozen`, `stream7-ca-frozen`, `stream7-cb-frozen`) before the verdict is computed. Any later change is a new prediction, not an edit.
3. **Disclose prior knowledge.** If the relevant bound was already known to the authors, the freeze is recorded as a *retrodiction*; only a freeze made before its comparison document was pinned is a *forward* prediction.
4. **Compute the verdict mechanically** in Lean against the pinned bounds, and record it whatever it is — including “not falsifiable in practice”, which by this protocol is not a test.
5. **No post-hoc conventions.** An order-one factor may not be chosen after the bounds are read; it must be derived, or the prediction is taken with the convention stated in the freeze.

Two of the four freezes (P1, C-A) are retrodictions: the bounds, and in C-A’s case the outcome, were known in advance, and the frozen documents say so. The C-B freeze preceded the pinning of the detector document it was compared with.

3 The programme’s one physical identification

The dual-scale programme pairs a micro length $\ell_{\text{micro}} = \ell_P$ with a macro length $\ell_{\text{macro}} = c/H_0$ as a T-dual pair with $\alpha' = \ell_{\text{micro}}\ell_{\text{macro}}$ **Tier C — conjecture**. Its self-dual length is $s = \sqrt{\alpha'}$. With $\ell_P = 1.616255 \times 10^{-35}$ m (CODATA 2018) and $c/H_0 = 1.3672 \times 10^{26}$ m ($H_0 = 67.66$ km/s/Mpc, Planck 2018 VI [6]) **Tier L — literature**, $s^2 \approx 2.2097 \times 10^{-9}$ m², $s \approx 47.0$ μm , and $(40 \mu\text{m})^2 \leq s^2 \leq (70 \mu\text{m})^2$ (`SelfDualCutoff.selfDual_length_sq_bracket`) **Tier A — Lean kernel**.

Three facts from paper 9 frame what follows.

- **CKN.** With $L = \ell_{\text{macro}}$, $M = 1/\ell_{\text{micro}}$ and UV length ℓ_{UV} , the corrected CKN bound holds iff $\ell_{\text{micro}}\ell_{\text{macro}} \leq \ell_{\text{UV}}^2$ (`SelfDualCutoff.ckn_iff_uvLength_ge_selfDual`) **Tier A — Lean kernel**; the literal hypothesis (Planck-scale cutoff with Hubble IR) fails by at least 10^{30} (`CKNInstance.planckEnergy_gt_ckn_horizon_cutoff`) **Tier A — Lean kernel**.
- **Regge reading excluded.** If $\alpha' = s^2$ is the Regge slope, string resonances sit near $\hbar c/s \approx 4$ meV; CMS excludes them below 7.9 TeV in low-string-scale models [7] **Tier L — literature**, and s^2 exceeds the CMS ceiling $(\hbar c/7.9 \text{ TeV})^2$ by at least 10^{30} (`CosmicString.selfDual_alphaPrime_exceeds_cms_ceiling`) **Tier A — Lean kernel**.

- **Dark-energy identity.** With $\rho_\Lambda = \Omega_\Lambda 3H_0^2/(8\pi G)$, $\rho_\Lambda^{-1} = (8\pi/3\Omega_\Lambda)(\ell_P L_H)^2$ (DarkEnergyScale.rhoLambda_inv_eq) **Tier A — Lean kernel**: the self-dual length *is* the dark-energy length up to $(8\pi/3\Omega_\Lambda)^{1/4}$ — an identity, not a corroboration.

4 Inventory: what can be derived?

Table 1 lists every output of the programme’s five formal streams and asks whether it is a number with units derivable without an unconstructed compactification.

Stream	Output	Observable?
1–2	lattices, $O(d, d; \mathbb{Z})$, double field theory, the dual-scale bound, flux/tadpole arithmetic	no — no number with units
3	self-dual length s	yes — P1 (Section 5)
3	CKN saturation with infrared length $c/H(t)$	yes, given a Tier C identification — C-A (Section 6)
3	scale-factor duality $a \leftrightarrow 1/a$	only with dilaton-gravity dynamics — C-B (Section 7)
3	cosmic F-string tension from $\alpha' = s^2$	yes — C-C (Section 8)
4–5	moonshine, umbral forms, dyon counts on $K3 \times T^2$	no — dimensionless, and properties of $\mathcal{N} = 4$ string theory

Table 1: Inventory of derivable observables (docs/STREAM7_HYPOTHESIS_INVENTORY.md).

Streams 4–5 deserve a word. Type IIA string theory on $K3 \times T^2$ gives an $\mathcal{N} = 4$ theory in four dimensions (Aspinwall, l. 2743 [13]) **Tier L — literature**; four-dimensional $\mathcal{N} = 4$ supersymmetry is non-chiral (l. 2715) and admits no chiral matter (standard background, **Tier L — literature**). The moonshine and dyon results [4] are therefore mathematics of a supersymmetric string compactification, not a model of our universe, and yield no observable here.

5 P1: the self-dual length as an extra dimension

Frozen statement (tag stream6-p1-frozen, commit 53f15b7; retrodiction, disclosed). One large extra dimension of radius $R = s$, with the Kaluza–Klein convention $m \sim 1/R$ used by Montero–Vafa–Valenzuela (MVV, $m^{-1} \sim l$, l. 327 [9]) **Tier C — conjecture**.

Tests fixed in advance **Tier L — literature**:

- T1: Eöt-Wash 2020, “the largest extra dimension must have a toroidal radius less than $30 \mu\text{m}$ ” (l. 289 [8]);
- T2: any gravitational-strength Yukawa interaction has range $\lambda < 38.6 \mu\text{m}$ (l. 285), read with $\lambda = R$ for the lowest KK mode (**Tier C — conjecture**);
- T3: heating of old neutron stars, one extra dimension, $l < 44 \mu\text{m}$ (MVV l. 327).

Outcome rule: *excluded* if T1 fails; *in tension* if only T2 or T3 fails; *consistent* otherwise.

Theorem 5.1 (Verdict on P1). **Tier A — Lean kernel** With $R^2 = \ell_P c/H_0$ (Stream6Verdict.p1RadiusSq): $(30 \mu\text{m})^2 < R^2$, $(38.6 \mu\text{m})^2 < R^2$ and $(44 \mu\text{m})^2 < R^2$ (p1_fails_T1, p1_fails_T2, p1_fails_T3; p1_verdict_excluded). Margins: $R > 1.5 \times 30 \mu\text{m}$ and $R > 1.06 \times 44 \mu\text{m}$ (p1_margin). Passing T1 would need $R = \kappa s$ with $\kappa < 0.64$ (p1_factor_needed: $\kappa = 0.64$ fails, $\kappa = 0.63$ passes).

By the frozen rule, **P1 is excluded**. The factor κ is recorded, not chosen: with $R = s/2\pi \approx 7.5 \mu\text{m}$ the number would fall inside MVV’s “dark dimension” window $0.1\text{--}10 \mu\text{m}$ (l. 427 [9]), and choosing that convention after reading the bounds is exactly what the protocol forbids.

Can κ be derived? Yes, and it is 1. Within the programme’s own T-duality $R \mapsto \alpha'/R$ with $\alpha' = s^2$:

Theorem 5.2 (The self-dual radius is fixed). Tier A — Lean kernel For $R, s > 0$: $R = s^2/R \iff R = s$ (*p62_selfdual_fixed_point*); at $R = s$ the Kaluza–Klein scale $1/R$ equals the winding scale R/α' (*p62_towers_coincide*).

With the standard normalisation $m_n = n/R$, the programme’s radius is $\sqrt{\alpha'} = s$: $\kappa = 1$. Taking both circles of T^2 large makes things worse: MVV’s two-dimension bound $l < 1.6 \times 10^{-4} \mu\text{m}$ (l. 329) is missed by more than 10^5 (*p62_two_dims_excluded*) Tier A — Lean kernel.

6 C-A: holographic dark energy at the Hubble scale

Frozen statement (tag *stream7-ca-frozen*, commit 90217a3; retrodiction with a foreseeable outcome, disclosed: Hubble-scale holographic dark energy is classically objected to because it does not accelerate, and the acceleration is observed). Assumptions Tier C — conjecture: (A1) $\rho_\Lambda(t) = \varepsilon 3H(t)^2/(8\pi G)$ with constant $0 < \varepsilon < 1$ (CKN saturation with infrared length c/H); (A2) flat FRW, $H^2 = (8\pi G/3)(\rho_m + \rho_\Lambda)$, with matter separately conserved, $\rho_m \propto a^{-3}$.

Theorem 6.1 (Derivation). Tier A — Lean kernel Write $k = 8\pi G/3$. (i) (A1)–(A2) force $\rho_m = (1 - \varepsilon)H^2/k$ (*Stream7CA.ca_matter_fraction*). (ii) Hence $H(a)^2 = C/a^3$ with $C = k\rho_{m0}/(1 - \varepsilon)$ (*ca_hubble_scaling*). (iii) For $E(a) = C/a^3$, $E'(a) = -3C/a^4$ (*hasDerivAt_E*) and the deceleration parameter $q = -1 - aE'(a)/(2E(a))$ equals $1/2$ at every $a > 0$ and for every $C > 0$ (*ca_deceleration*).

The prediction has no free parameter: whatever ε , the universe decelerates with $q = +1/2$.

Test and verdict. Rule fixed in advance: C-A passes iff $q_0 \geq 0$. With Planck 2018’s $\Omega_\Lambda = 0.68885$ in flat ΛCDM (*DarkEnergyScale.omegaLambda*) Tier L — literature, $q_0 = (1 - \Omega_\Lambda)/2 - \Omega_\Lambda \approx -0.53 < 0$, and $q_0 \neq 1/2$ (*ca_verdict_excluded*) Tier A — Lean kernel. **C-A is excluded.** The input q_0 is model-dependent (a ΛCDM fit), as recorded in the freeze; the qualitative fact tested — present acceleration — does not rest on ΛCDM alone, but no model-independent source is pinned here.

7 C-B: pre-big-bang relic gravitons at the programme’s string scale

Frozen statement (tag *stream7-cb-frozen*, commit 60320a4; frozen *before* any detector document was pinned). Assumptions Tier C — conjecture: (B1) the programme’s scale-factor duality is realised dynamically as a pre-big-bang phase; (B2) $\alpha' = s^2$ fixes the string scale, so the transition curvature $g_1 = H_1/M_P \simeq M_s/M_P = \ell_P/s$.

Literature input Tier L — literature (Gasperini–Veneziano [10]). For $a = |\eta|^\alpha$ the relic spectrum is $\Omega \sim \omega^{3-2\nu}$, $\nu = |\alpha - 1/2|$ (Table 2, ll. 4467–4485); the dilaton-driven phase has $\alpha = 1/2$, hence $\Omega \sim \omega^3$ (l. 4505) in the low band (5.15), ll. 5040–5053; end points $\omega_1 \simeq g_1^{1/2} 10^{11}$ Hz and $\Omega(\omega_1) \simeq 10^{-4}g_1^2$, with $g_1 = H_1/M_P \simeq M_s/M_P$ ((5.16), ll. 5058–5061).

Theorem 7.1 (Derived numbers). Tier A — Lean kernel *The slope $3 - 2|\alpha - 1/2|$ is 3 at $\alpha = 1/2$ and 0 (flat) at $\alpha = -1$ (`Stream7CB.cb_slope`). With $g_1^2 = \ell_P/(c/H_0)$ (`g1Sq`): $1.1 \times 10^{-61} \leq g_1^2 \leq 1.3 \times 10^{-61}$ (`cb_g1_sq_bracket`); $\omega_1 < 10^{-4}$ Hz, stated as $\omega_1^4 = g_1^2 \cdot 10^{44} < 10^{-16}$ (`cb_peak_below_lisa_band`); the peak $\Omega(\omega_1) \simeq 10^{-4}g_1^2$ (`omegaPeak`) is below 10^{-64} (`cb_peak_amplitude_unreachable`).*

So $\omega_1 \approx 6 \times 10^{-5}$ Hz and $\Omega_{\text{GW}}(\omega_1) \approx 1.2 \times 10^{-65}$; the only free element, the break $\omega_s \leq \omega_1$, can only lower the spectrum.

Test and verdict. Rule fixed in advance: testable iff the prediction reaches the design sensitivity of the most sensitive detector whose design document is pinned after the freeze. The LISA mission proposal requires the ability to measure $\Omega = 1.3 \times 10^{-11}(f/10^{-4} \text{ Hz})^{-1}$ for $0.1 < f < 2$ mHz (OR7.2, ll. 692–699 [11]) Tier L — literature, i.e. at best $\approx 6.5 \times 10^{-13}$ at 2 mHz (`omegaLisaBest`). The predicted spectrum ends below LISA’s band and its peak, even multiplied by 10^{50} , stays below LISA’s best (`cb_not_testable`, `cb_peak_amplitude_unreachable`) Tier A — Lean kernel. **C-B is not falsifiable in practice** — neither confirmed nor excluded by its rule.

Stronger. Premise (B2) is the Regge reading of $\alpha' = s^2$, a meV string scale, which is already excluded (Section 3; `CosmicString.selfDual_alphaPrime_exceeds_cms_ceiling`). C-B fails at the level of its assumption, not only for lack of sensitivity.

8 C-C: cosmic F-strings

An unwarped cosmic F-string has $G\mu/c^2 = \ell_P^2/(2\pi\alpha')$ (Copeland–Myers–Polchinski [12]) Tier L — literature. For $\alpha' = \ell_P L$ this is $\ell_P/(2\pi L)$ (`CosmicString.fStringGmu_selfDual`) Tier A — Lean kernel; with $L = c/H_0$, $G\mu \approx 1.9 \times 10^{-62}$, and in any case $G\mu \leq 10^{-61}$, more than fifty orders below the lower end 10^{-11} of the brane-inflation window (`CosmicString.selfDual_Gmu_below_cmp_window`) Tier A — Lean kernel. A prediction that no experiment can reach is not a test; C-C is recorded as out of reach (paper 9 deliberately does not count the Planck $G\mu$ bound, passed by 55 orders, as evidence). It shares premise (B2) with C-B.

9 The common cause

Observable	Freeze	Verdict	Where $\alpha' = \ell_P c/H_0$ enters
P1	retrodiction	excluded (T1–T3)	$R = \sqrt{\alpha'}$, fixed by T-duality
C-A	retrodiction	excluded ($q = +1/2$ vs $q_0 < 0$)	IR length $c/H = \ell_{\text{macro}}$
C-B	forward	not falsifiable in practice; premise excluded	$M_s \sim 1/\sqrt{\alpha'} \sim 4$ meV
C-C	(paper 9)	out of reach; premise excluded	$G\mu = \ell_P^2/(2\pi\alpha')$

Table 2: The four verdicts and the role of the one identification.

Each verdict traces back to the identification $\alpha' = \ell_{\text{micro}}\ell_{\text{macro}} = \ell_P c/H_0$ (Table 2). Read as a length, it lands at tens of microns, where gravity has been tested and it conflicts with data (P1). Read as an energy density at the Hubble scale, it fixes the dynamics with no room for acceleration (C-A). Read as a string scale, it is a meV scale: excluded by colliders and, as a by-product, hiding every stringy relic below any sensitivity (C-B, C-C). Together with the CKN result, the dual-scale hypothesis in all of its testable forms does not survive the existing data.

10 What survives

The verdicts concern physical identifications (Tier C). The Tier A mathematics of the programme is untouched: lattices, $O(d, d; \mathbb{Z})$ and the dual-scale bound $\text{tr } G + \text{tr } G^{-1} \geq 2d$ with its unique minimiser (`dualScale_eq_iff`, paper 8 [2]); Mathieu moonshine computed at all 26 classes and levels 0–9 (`moonshine_modules_decompose`), the shadow’s 24 (`shadow_coeff_eq_perm_trace`), and the single-centred dyon counts on $K3 \times T^2$ given by Hurwitz class numbers (`immortal_m1`), all in paper 10 [4]. The same twining (“forger’s”) test that the number $24 = \chi(K3)$ passes is failed by paper 7’s “27720 lock” at every non-identity class (`ratio_fails_at_every_class`); paper 7’s Revision 5 records this.

What is not proved

That any of the identifications (P1, A1–A2, B1–B2) is part of string theory (Tier C); that the Tier L inputs are correct beyond their sources; a model-independent measurement of q_0 (not pinned); any realistic compactification. Two of four freezes are retrodictions, as disclosed. “Not falsifiable in practice” is not a confirmation.

11 Lessons for pre-registered theory testing

1. **Inventory before hypothesis.** Asking first which observables are derivable at all prevented a search for a convenient one; the answer (four) bounded the whole exercise.
2. **Freeze the order-one factors.** Paper 9’s “disfavoured by $O(1)$ ” became an exclusion once the factor was either fixed in the freeze or derived (Theorem 5.2). An unfixed factor is where post-hoc rescue lives.
3. **Retrodiction is still worth recording.** Even with a known outcome, a frozen, kernel-checked derivation (C-A) turns an informal objection into a checkable one and removes it from future debate.
4. **Separate “not excluded” from “not testable”.** C-B and C-C are neither confirmed nor excluded by their rules; the protocol says so explicitly rather than counting consistency with an unreachable bound as support.
5. **Name quantities in verdict theorems.** A review of release v3.16.0 found two C-B statements whose second conjunct was a fact about powers of ten; they were restated with named quantities (`omegaPeak`, `omegaLisaBest`), with the statement lock recording exactly those two changes.

12 Open directions

1. **Drop or replace $\alpha' = \ell_{PC}/H_0$.** Any new hypothesis worth testing must do without this identification; its observables must again be derived and frozen before comparison.
2. **A chiral geometry.** $K3 \times T^2$ gives $\mathcal{N} = 4$ in four dimensions; a realistic model needs a geometry with a chiral spectrum — for instance a $K3$ -fibred Calabi–Yau threefold — together with an explicit construction (fluxes, moduli stabilisation). This is new research, not a patch.
3. **Model-independent inputs.** Pinning a model-independent determination of the present acceleration would lift C-A’s verdict from a Λ CDM-conditional one.
4. **Reusable protocol.** The freeze-by-tag protocol and the Lean verdict files are generic and could be reused by other programmes that make order-one-sensitive claims.

13 Conclusion

A hypothesis that links the Planck length and the Hubble radius through string T-duality yields exactly four observables without an unconstructed compactification. Frozen before their verdicts and checked in Lean against pinned data, two are excluded and two are out of reach with an excluded premise. The single cause is the identification $\alpha' = \ell_{PC}/H_0$. We regard this as a clean negative result: the programme’s testable physical content does not survive, its mathematics does, and the protocol that established this is reusable.

A Reproducibility

Build and audit. `lake build DualScaleCosmology`

`python3 tools/axiom_audit.py DualScaleCosmology`

reports 50 theorems, 0 failing; across the ten libraries of the repository, 621 theorems, 0 failing (release v3.16.1). Axioms used: at most `propext`, `Classical.choice`, `Quot.sound`.

Statement **lock.** `python3 tools/statement_lock.py -check $(find DualScaleCosmology -name '*.lean')`

locks 68 declarations in 10 files, including `Stream6Verdict.lean`, `Stream7CA.lean` and `Stream7CB.lean`.

Freezes. `git show stream6-p1-frozen, stream7-ca-frozen, stream7-cb-frozen` (commits 53f15b7, 90217a3, 60320a4) display the frozen documents; the verdict files were committed afterwards.

Sources pinned (hash-registered in `papers/foundations/MANIFEST.md`): 2002.11761, 2205.12293, `hep-th/0207130`, 1702.00786, `hep-th/0312067`, 1911.03947, `hep-th/9803132`, `hep-th/9611137`; CODATA 2018 and Planck 2018 VI constants via `astropy`.

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